



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Food Surplus Management and Redistribution System

Team ID: DSCPP-III-2026-T074

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27



Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T074

Semester: 3rd

Section: ML 1, ML 2, ML 4, DS 1

Domain: Data Structures and Object Oriented Programming in C++

Team Members

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Mentor

Mr. Kartikey Arora

Interactions completed: 4



Problem Statement & Motivation

Problem Statement

Every day, large quantities of usable food are wasted at various events whereas a lot of people still face food insecurity. This project aims to solve this problem of food insecurity and wastage by timely coordination and redistribution of surplus food to those in need.

Motivation

- ▶ Lack of timely coordination
- ▶ This may lead to food expiry before redistribution
- ▶ Lack food expiry and safety standards

Target Users / Stakeholders

- ▶ Primary Users: Donors and recipients
- ▶ Secondary Users: None as in Phase 1
- ▶ Benefit- Timely, expiry/urgency based redistribution of surplus food.

Objectives

Project Objectives

1. Develop a system to manage and redistribute usable surplus food before expiry.
2. Implement registration, validation, requests and basic matching according to urgency and food expiry window.
3. Integrate C++ Object Oriented Programming with core Data Structures.
4. Evaluate validation, expiry, allergen, duplicate records and matching under various scenarios.
5. Demonstrate Phase 1 workflow and future scalability.

Key Objective: To develop a system for timely management and redistribution of surplus food based on food availability, expiry and basic requirements.

Proposed Solution

Proposed Approach

1. **Input / Data:** Donor and recipient details, food information.
2. **Processing:** Data Validation → Expiry/Urgency Check → Food Prioritization
3. **System Logic:** C++ OOP engine using linked list, min-heap, hash table & FIFO queue.
4. **Storage:** C++ File I/O for user, food, and request data.
5. **Output:** Food booking and donor, recipient notification.

Key Features

- ▶ Role-based donor/recipient UI
- ▶ Food registration & validation
- ▶ Min-Heap urgency priority
- ▶ Hash Table ID/allergen lookup
- ▶ FIFO requests & basic matching

The system enables structured and timely redistribution of surplus food by considering expiry, quantity, allergens, dietary requirements, and recipient needs.

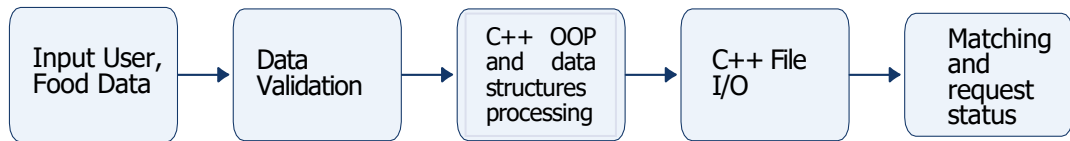


Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Linked Lists, FIFO Queue, Min Heap, Hash Table.	Food Inventory, Urgency Prioritization & Request Management
OOPs with C++	Classes & Objects, Inheritance, Polymorphism, Encapsulation, Abstraction.	User, food and request management.
Operating Systems	Not used.	None
DBMS	Not used. C++ File I/O	User, Food & Request Data Storage

Requirement: Integration of OOP-based entity management with data structures for inventory organization, expiry-based urgency prioritization, efficient searching, and recipient-food matching.

System Architecture / Workflow



Major Components

- ▶ Frontend: User Interface
- ▶ C++ Core: OOPs, Data Structures & File I/O
- ▶ Output Module: Food Recommendations & Requests

Data /Control Flow

- ▶ **Data Flow:** User/Food data → Data Validation → OOPs & Data Structures → File I/O → Matching & Status Display
- ▶ **Module Interaction:** Frontend ↔ C++ OOPs ↔ Data Structures ↔ File I/O
- ▶ **Output:** Food recommendations and redistribution requests.



Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C++, HTML, CSS, JavaScript*
- ▶ Storage: *C++ File I/O*
- ▶ Tools / Framework: *C++ STL*
- ▶ IDE: *VisualStudio, Gcc*
- ▶ Version Control: *GitHub*

Development Methodology

1. Requirement Analysis: Analyze food wastage and user needs
2. System Design: Workflow and data structures
3. Implementation: C++ OOPs, Data Structures, and File I/O
4. Frontend Integration: User interface and input handling
5. Testing and Evaluation: Validation, expiry, and matching

C++ supports OOPs and data structures used to build the backend, File I/O provides data storage, HTML/CSS/JavaScript creates the user interface. The iterative methodology supports gradual development, integration, and testing.

Initial Progress & Team Contribution

Progress Achieved

- ▶ Background: Food wastage & redistribution systems study.
- ▶ Requirements: User, food, and matching requirements.
- ▶ Architecture: Frontend, C++ Core and File I/O workflow
- ▶ Initial module: Frontend & C++ OOPs/DS modules
- ▶ Data preparation: C++ File I/O and food data records

Individual Contribution

Member	Role	Contribution
M1	Team Lead	Backend development, Coordination and documentation
M2	Developer	Frontend Development, Integration
M3	Developer	Backend Development
M4	File I/O, Testing	Data Storage, Testing/Debugging



Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ Phase-I frontend with UI
- ▶ Validation, expiry and matching tests
- ▶ Food redistribution and management
- ▶ Future scalability- Integration with AI/ML

References

1. Cattaneo, A., Sánchez, M. V., Torero, M., & Vos, R. (2021). Reducing Food Loss and Waste: Five Challenges for Policy and Research. Food Policy, 98, 101974. <https://doi.org/10.1016/j.foodpol.2020.101974>
2. The C++ Programming Language — Bjarne Stroustrup
3. MDN Web Docs and cppreference
4. GeeksforGeeks C++ Data Structures resources

Thank You
Questions & Suggestions